Learning Objectives 6.1–6.3: I can calculate the total current through a surface area given a current distribution. I can use Ohm's Law for electromagnetics to calculate the relationship between a current distribution and an electric field. I can articulate and identify what makes a material a good conductor and how conductivity relates to resistance.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). Use $e = 1.602 \times 10^{-19}$ C and $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

1. **Current through a surface.** A wire of radius $a = 2$ mm lies along the z axis and carries the current density $\mathbf{J} = \hat{\mathbf{z}}\, J_0 \left(1 - \rho^2/a^2\right)$, with $J_0 = 5 \times 10^5$ A/m^2.

   (a) Charge of density ρ_v moves with velocity $\mathbf{u}$. Show that the current through a small area ΔS with unit normal $\hat{\mathbf{n}}$ is $\Delta I = (\rho_v \mathbf{u}) \cdot \hat{\mathbf{n}}\, \Delta S$.

   (b) Find the total current in the wire.

   $I =$ ☐

   (c) Find the current inside $\rho < a/2$, and the fraction of the total it carries.

   $I_{1/2} =$ ☐

   fraction = ☐

2. **Ohm's law in point form.** An aluminum wire of cross section 2.5 mm^2 carries 10 A uniformly. Aluminum has $\sigma = 3.7 \times 10^7$ S/m and $n = 1.8 \times 10^{29}$ free electrons per cubic meter.

   (a) Find the current density, the field in the wire, and the voltage across a 20 m length.

   J =

   E =

   V =

   (b) Find the drift speed of the electrons and their mobility.

   u =

   μ_e =

   (c) Aluminum has more free electrons per cubic meter than copper (8.5×10^{28}), yet a lower conductivity. Explain why, using $\sigma = ne\mu_e$.

   (d) A power line must have the same resistance per meter in aluminum as in copper ($\sigma = 5.8 \times 10^7$ S/m). Find the ratio of the cross sections, and the ratio of the masses per meter. The densities are 2700 and 8960 kg/m^3.

   area ratio =

   mass ratio =

3. **Resistance of a wire.** A straight wire of length ℓ, cross section A, and conductivity σ has its end at $z = 0$ held at voltage V and its end at $z = \ell$ at zero.

   (a) Assume a uniform field $\mathbf{E} = \hat{\mathbf{z}}\, E$ in the wire. Integrate for the voltage and the current, and derive $R = \ell/(\sigma A)$.

   (b) Explain why the field must have the same magnitude all along the wire when the current is steady.

   (c) Find the DC resistance of 100 m of copper wire of diameter 2.05 mm.

   R =

4. **Leakage through a coaxial line.** A coaxial line has inner radius a, shield radius b, and a dielectric of permittivity ϵ and small conductivity σ.

   (a) Place ρ_ℓ per meter on the inner conductor. Use the field from Gauss's law to derive the leakage conductance per meter, $G' = 2\pi\sigma/\ln(b/a)$.

   (b) Moisture raises the conductivity of the dielectric in a 50 m run of RG-58 ($a = 0.405$ mm, $b = 1.475$ mm, $\epsilon_r = 2.25$) to 10^{-12} S/m. Find G' and the DC resistance between the conductors.

   G' = ____

   R = ____

   (c) Lesson 25 gives $C' = 96.8$ pF/m for this cable. Show that $G'/C' = \sigma/\epsilon$ with your numbers.

Documentation: _____